



PLATFORM TECHNOLOGIES

Chrisna L. Fuco



Bachelor of Science in Information Technology
College of Computing Studies

v2.01182024

Table of Contents

Module 5: Overview of Technology Platform

Introduction	76
Learning Outcomes	76
Lesson 1. What is Technology Platform?	76
Lesson 2. Benefits of Modern Platform	77
Lesson 3. When do you need a cloud technology platform?	78
Lesson 4. Technology platform components	79
Lesson 5. Types of Technology Platforms	81
Assessment Task	87
Summary	89
References	90

Module 6: Database Concept, Software and Cloud Containers

Introduction	92
Learning Outcomes	92
Lesson 1. What is Database?	92
Lesson 2. Types of Databases	95
Lesson 3. Front-end and Back-end Development	96
Lesson 4. Cloud infrastructure	97
Assessment Task	98
Summary	99
References	99

Module 7: Types of Software Platforms

Introduction	100
Learning Outcomes	101
Lesson 1. Utility Platforms	101
Lesson 1.1 What is utility platform?	101
Lesson 1.2 Utility platform in various domain	102
Lesson 2. Content Distribution Platforms	106
Lesson 2.1 What is content distribution platform?	106
Lesson 2.2 Best content distribution platform	107
Assessment Task	120
Summary	121
References	123

MODULE 5

OVERVIEW OF TECHNOLOGY PLATFORM



Introduction

Since the advent of computer technology, business applications have provided organizations with the tools to overcome common business challenges. Over decades, applications have radically changed, while the following has remained consistent: business apps are more often purchased based on functionality rather than the underlying technology. But what's become clear is that legacy applications that lack the support of a robust technology platform often create unanticipated problems. When it comes to your business-critical applications, the underlying technology platform is now more critical than ever.



Learning Outcomes

At the end of this lesson, the student should be able to:

1. Discuss the different platform technology in computing.
2. Demonstrate basic cloud computing using different platforms, like mobile, cloud, or web containers.

Lesson 1. What is Technology Platform? (www.sap.com)

A **technology platform** is the foundation for building and running business applications. The platform allows users to run their applications smoothly without worrying about the technology that supports them. At the same time, it allows technical staff to rapidly extend, enhance, or upgrade application software, increasing the speed of business.

A modern technology platform typically includes analytics, database and data management, tools for application development and extension, integration, and intelligent technologies such as artificial intelligence (AI), machine learning, and the Internet of Things (IoT).

Lesson 2. Benefits of a modern platform (www.sap.com)

A modern technology platform is the key driver that allows business to scale, integrate, and extend their business applications – critical elements when it comes to business transformation.

Here are some of the common benefits of a modern technology platform:

- Supports scaling with business growth
- Provides data management
- Supports workload management
- Connects and accelerates development through internal and third-party integration
- Provides a modern user interface (UI) and user experience (UX)
- Meets the needs of mobile workers
- Supports new, specialized applications, such as logistics and tax cloud services
- Provides standard productivity tools such as alerts, KPIs, workflows, drill down, and self-service reporting
- Secures business data

A modern technology platform provides benefits beyond traditional application support. Today's technology platform is critical in digital transformation, providing functionality that improves processes and drives business value for customers. Digital transformation may look different for every organization, but it will always affect all areas of a company – introducing critical changes to the way business units operate.

Modern digital transformation demands a technology stack that can meet the ever-changing business landscape. Below are some new, essential capabilities to help facilitate innovation:

- Artificial intelligence technologies, such as machine learning, robotic process automation (RPA), and IoT
- Advanced analytics for improved decision-making
- Simplified component building blocks for lower development and maintenance costs
- Faster application development through the use of low-code and no-code tools for both pro and citizen developers
- A personalized experience for today's modern workforce
- Modernized databases for both transactional and analytical workloads
- Out-of-the-box integration between popular applications like ERP, CRM, HRM, and procurement systems from many different and varied vendors

Lesson 3. When do you need a cloud technology platform?

(www.sap.com)

There are many reasons why a company will decide that they need a better, cloud-based technology platform. The following are five of the frequently cited reasons people make the switch:

1. **They need to be futureproof.** When new business functionality and technology break into the marketplace, legacy technology stacks cannot support it. With a cloud platform, companies can adopt what they need, when they need it – to stay on the cutting edge.
2. **They are expecting enterprise volume and/or Big Data growth.** A fast-growing company expects a significant increase in transaction volume and the growth of Big Data, making moving to a scalable cloud platform a wise overall business decision.
3. **They want to lower the total cost of ownership.** Old technology, both hardware and software, requires a great deal of maintenance and very skilled technical people that may or may not be available or cost-effective. With a cloud platform, the vendor often takes care of the maintenance for you, freeing up costs and your staff to work on the growth of your business.

4. **They want to take advantage of cloud capabilities.** As a result of new and expanding capabilities, there is now significant momentum behind cloud computing deployment. One of the big attractions is a modern cloud technology platform with elasticity, high performance, security, and privacy.
5. **They want to ensure a secure, compliant cloud solution.** Security concerns rank top in many surveys involving cloud adoption and ensuring data, integration, and all access points are secure is key to strengthening against hackers – also to maintaining compliance with local laws and enabling sophisticated access security for employees, customers, and partners.

Lesson 4. Technology platform components (www.sap.com)

The modern technology platform has many different possible components. These components can be anything from products designed for specific functionality to individual platforms that can stand alone or be part of an integrated package. These components or layers are vital for a modern, cloud-based technology platform.

- **User Interaction** - The user interaction is made up of two parts: the user interface (UI) and the user experience (UX). The UI is the visual and manual way the user interacts with a system or machine. For example, a business application includes the layout of the screen, colors, icons, shortcuts, and more. It also includes mobile, mouse, voice, or gesture input. The UX is a user's personal experience when using a system. The design's degree of intuitiveness and user-friendliness will significantly impact the user's productivity.
- **Data insight** - Data insights have become a driving reason to move to modern ERP on a modern tech platform. Businesses today need to have accurate, real-time information at the decision-maker's fingertips. Unfortunately, companies found that the information they were getting from their legacy systems was out-of-date and incorrect.

- **Application** - The application layer provides functionality for performing business tasks. Applications can span the enterprise for example through ERP or CRM or address a narrower scope, such as departmental accounts payable. At a glance, the software providers behind the applications may look similar, but the application development platforms can be radically different underneath. It is this difference that significantly impacts how the system performs and is updated.
- **Integration** - The integration may be simple, such as an extension of the core functionality with third-party features and functions or industry-specific add-ons designed to work seamlessly with the primary system. In other cases, the work could be more complicated, involving integration with another ERP or CRM application or a customer's proprietary systems. The ease of the initial integration and maintenance depends on the application programming interfaces (APIs).
- **Database** - Data storage and management have always been critical components of a technology platform. Legacy ERP systems worked well in the past but cannot keep up with the current explosion of Big Data. In-memory databases provide the speed that is required, but until recently, they have been expensive. However, thanks to new technological advancements, the cost of in-memory databases has lowered and has become a more viable alternative for large systems.
- **Deployment** - This final layer must support all deployment options – including cloud, on-premise, hybrid, and edge computing. Cloud platforms are now the most popular choice for new ERP functionality. Not only do they require minimal hardware management by the customer, but much of the deployment of applications is also run in secure, virtual instances for the customer, including data backup and additional resources when required.

Lesson 5. Types of Technology Platforms (www.sap.com)

A technology platform is a **structure on which users build or run successful business applications**. It acts as a building block for extending and developing an application, driving innovation, and fostering business growth.

TYPES OF TECHNOLOGY PLATFORMS

1. Operating System

- ✓ An **operating system** is a program that controls the execution of application programs and acts as an interface between the user of a computer and the computer hardware.
- ✓ A more common definition is that the operating system is the one program running at all times on the computer (usually called the kernel), with all else being application programs.
- ✓ An **operating system** is concerned with the allocation of resources and services, such as memory, processors, devices, and information. The operating system correspondingly includes programs to manage these resources, such as a traffic controller, a scheduler, a memory management module, I/O programs, and a file system.

Functions of Operating System – The operating system performs three functions:

1. **Convenience:** An OS makes a computer more convenient to use.
2. **Efficiency:** An OS allows computer system resources to be used in an efficient manner.
3. **Ability to Evolve:** An OS should be constructed in such a way as to permit the effective development, testing, and introduction of new system functions at the same time without interfering with service.

Examples of Operating Systems

- Microsoft Windows.
- macOS.
- Linux.
- Android.
- Apple iOS.

2. Computing Platforms

- ✓ A **computing platform** consists of hardware and an operating system on which you can run an application or a program.
- ✓ Although computing platforms exist in different forms, you need to define the system and its underlying requirements and limitations.
- ✓ Ultimately, all computing platforms have the same general hardware, including a processor, operating memory, and storage system. Some come with input or output devices that aid user interactions.

Examples of Cloud Platforms

- A personal desktop computer using Windows OS.
- An iPad running on the iOS system.
- A minicomputer utilizing the Linux OS alongside other variations.

3. Database Platforms

- ✓ Database platforms are **cloud platforms that enable you to deploy and manage different database types** like relational, NoSQL, and in-memory databases.
- ✓ A database management system (DBMS) stands between end users, the database, and applications to help capture and analyze the data shared between these mediums.

Examples of Database Platforms

- Oracle
- IBM Db2
- Microsoft SQL
- MySQL
- PostgreSQL
- MongoDB

4. Storage Platforms

- ✓ Storage platforms allow the storage of files and objects. There are different storage platform types: object-based storage, software-defined storage, and data-defined storage.
- ✓ Object storage manages unstructured data as objects. Software-defined storage separates storage services from physical storage hardware. Data-defined storage

combines the attributes of software and object-defined storage while adding data security and identity management.

Examples of Storage Platforms

- Google Drive.
- Dropbox.
- Onedrive.
- MediaFire.
- iDrive.
- Amazon Cloud Drive.

5. Application Platforms

- ✓ An application platform is a virtual platform that application programs rely on to conduct their standard operations. It provides an enabling environment for developing application software.

Examples of Application Platforms

- OutSystems
- Appery.io
- Microsoft Visual Studio
- Adobe PhoneGap
- Google App Maker

6. Mobile Platforms

- ✓ A mobile platform combines software tools to create, design, and maintain mobile applications. It includes operating systems and environments that enable the development of mobile apps.
- ✓ Every mobile platform supports the development of mobile apps by using different programming languages and an application programming interface that encourages interactions between software packages.

Examples of Mobile Platforms

- BlackBerry.
- iOS.
- Android.

- Windows Mobile OS

7. Web Platforms

- ✓ Web platforms allow for the discovery and sharing of information over the internet. They include web servers, web application servers, and digital elements that allow for the sharing of information online.
- ✓ A website that provides historical information about the United States of America is a great example of a web platform. Users can connect via the internet, type in the URL, and access the needed information.

Examples of Web Platforms

- Britannica.
- ResearchGate.
- Lexis Web.
- Wolfram Alpha.

8. Content Management System Platforms

- ✓ A Content Management System (CMS) platform **helps users to create, manage, and publish content, media, and documents**. This platform allows you to manage your content better and publish website content.
- ✓ Companies use CMS platforms as a document management system, an intranet site, and to publish web content on their blogs and websites.
- ✓ Without a CMS platform, you must first understand programming languages and coding properly. Why? All web pages use HTML, JavaScript, and CSS programming languages.
- ✓ The main benefit you gain from a CMS platform is **the ability to create, edit, and store web content without needing technical expertise**.

Examples of Content Management System Platforms

- Joomla.
- HubSpot CMS Hub.
- WordPress.org.
- WooCommerce.
- Drupal.
- BigCommerce.

- Shopify.

9. Media Platforms

- ✓ The primary function of a media platform is to provide and deliver media to its users. However, most media platform owners are now allowing feedback from their users to make their platforms more interactive.

Examples of Media Platforms

- Facebook
- WhatsApp
- LinkedIn
- YouTube
- Local Newspaper
- Network TV
- Pinterest

10. API Platforms

- ✓ Application Programming Interface (API) is an intermediary software providing the means for different applications to communicate with each other.
- ✓ An API platform allows for the easy management of APIs and other related applications. This platform exposes data, its resources, and assets in a machine-readable format to other users.

Examples of API Platforms

- Funnel.
- Make.
- Workato.
- Cyclr.
- DreamFactory.
- Maestro PMS

11. Analytics Platforms

- ✓ An analytics platform refers to any service or tool that allows users to capture, process, analyze, and visualize data for better decision-making. Organizations and different

departments (strategic business units) use analytics platforms to make data-driven decisions.

- ✓ Analytics platforms allow companies to employ their platforms on-premises or in the cloud. Industries such as marketing, finance, management, information security, online systems, and software services are the most significant users of analytics platforms.

Examples of Analytics Platforms

- Microsoft Power BI Desktop.
- Qlik Sense.
- Looker.
- Sisense.
- Tableau.

12. Security Management System Platforms

- ✓ Security management system platforms are systems that provide security teams with a centralized hub for managing their network security. These platforms integrate with third-party security products and services for better and faster outcomes.

Examples of Security Management System Platforms

- Quantum.
- CloudGuard.
- Harmony.
- Horizon.

13. Robotics Platforms

- ✓ Robotic platforms are systems or tools that provide the framework for developing, managing, and using robotic programs or devices.
- ✓ A robotic platform makes it easy for people with little or no technical expertise to create robotic programs and devices.

Examples of Robotics Platforms

- Google ROBEL.
- Microsoft AirSim.
- Apollo Baidu.

- iRobot AWARE
- OpenJAUS.
- Yarp.
- Pyro.

14. Internet of Things Platforms

- ✓ The Internet of Things (IoT) is a network of interconnected physical objects with sensors and processing abilities that collect and share data through the Internet.
- ✓ IoT platforms are on-premise software or cloud services that bridge the gap between device sensors and data networks.
- ✓ An IoT platform is a set of technologies that allow developers to build software applications, remotely collect data, and manage device connectivity. This platform connects different sets of technologies or communications to ensure a seamless flow of data between devices.

Examples of Internet of Things Platforms

- Google Cloud IoT.
- Salesforce IoT Cloud.
- IBM Watson IoT.
- Amazon AWS IoT Core.
- Microsoft Azure IoT Hub.
- Oracle IoT.

15. AI Platforms

- ✓ Artificial Intelligence (AI) platforms offer AI-based services that enable you to build and manage your own AI.
- ✓ AI platforms provide an enabling environment for machine learning, such as a machine learning database that enables you to build your AI applications effectively.
- ✓ As an integrated set of technologies, AI platforms enable organizations to create, develop, and operate their AI applications at scale.

Examples of AI Platforms

- Google AI Platform.
- TensorFlow.
- Microsoft Azure.

- Rainbird.
- Infosys Nia.
- Wipro HOLMES.

16. Gaming Platforms

- ✓ A gaming platform combines electronic or computer hardware and software to allow a video game to run efficiently. This platform is the supporting system that allows users to play multiplayer online games on different devices.
- ✓ The three most popular gaming platforms are personal computers, video game consoles, and mobile devices.

Examples of Gaming Platforms

- The Sony PlayStation.
- Microsoft's Xbox.
- Nintendo's Switch.
- PCs.
- Mobile.



Assessment Task

Instruction: Read the questions carefully and choose your answer in the box below.

a. Gaming Platforms	i. Media Platforms
b. AI Platforms	j. Content Management System Platforms
c. Internet of Things Platforms	k. Storage Platforms
d. API Platforms	l. Database Platforms
e. Robotics Platforms	m. Application Platforms
f. Security Management System Platforms	n. Cloud Platforms
g. Analytics Platforms	o. Operating System
h. Mobile Platforms	p. Web Platforms

- _____ 1. It is a platform that offers AI-based services that enable you to build and manage your own AI.
- _____ 2. This platform makes it easy for people with little or no technical expertise to create robotic programs and devices.
- _____ 3. It is an intermediary software providing the means for different applications to communicate with each other.
- _____ 4. This platform provides and delivers media to its users.
- _____ 5. It is a network of interconnected physical objects with sensors and processing abilities that collect and share data through the Internet.

Check for Understanding

Instruction: Match column A with the corresponding item in column B. Write the letter of your answer for each item.

_____ 1. Google AI Platform	a. Gaming Platforms
_____ 2. An iPad running on the iOS system	b. AI Platforms
_____ 3. Joomla	c. Internet of Things Platforms
_____ 4. WordPress.org	d. API Platforms
_____ 5. Harmony	e. Robotics Platforms
_____ 6. Infosys Nia	f. Security Management System Platforms
_____ 7. Qlik Sense	g. Analytics Platforms
_____ 8. Yarp	h. Mobile Platforms
_____ 9. WhatsApp	i. Media Platforms
_____ 10. Britannica	j. Content Management System Platforms
_____ 11. Microsoft Visual Studio	k. Storage Platforms
_____ 12. BlackBerry	l. Database Platforms
_____ 13. iOS	m. Application Platforms
_____ 14. Microsoft Windows	n. Cloud Platforms
_____ 15. Microsoft SQL	o. Operating System
	p. Web Platforms



A technology platform is the foundation for building and running business applications. Technology Platforms provide building blocks or services that are reused during sizable amount of products. Technology Platforms are not two-sided markets. Technology Platforms generate revenue by selling their services to developers and are typically invisible to top-level users (end users). For instance, while OTT's like Netflix and Amazon Prime run its video streaming services on top of Amazon Web Services platform (AWS), top-level users interact solely with Netflix and Amazon Prime. In this type of platform, there are no implicit network effects. These platforms grow with favorable reception by developers and do not rely on interaction between demand side and supply side. As a result, Technology Platforms are much easier to launch than multi-sided or peer-to-peer platforms. (<https://www.geeksforgeeks.org/types-of-software-platforms/>)

1. What is a platform?

- a group of technologies that are used as a base upon which other applications, processes, or technologies are developed.

2. Types of technology platforms?

- Database management system (DBMS), Engagement platforms, Digital marketplace, On-demand service platforms, Web platform, etc.,

3. What is an example of a technology platform?

- Amazon Web Services, Microsoft Azure, and Twilio are examples of Technology Platforms. Technology Platforms provide building blocks or services that are reused in a large number of products.

4. What are the 6 online platform categories?

- Learning Destination Sites.
- Traditional Commercial Learning Management Systems.

- Open Source Learning Management Systems.
- Modern Learning Management Solutions.
- Learning Management Ecosystems.
- Custom Built Platforms.



References

- a. Chakraborty, Pranabananda; *“Operating systems: evolutionary concepts and modern design principles”* CRC Press, 2024
- b. Rathee, Priyanka: *“Basic principles of an operating system: learn the internals and design principles”*. BPB Publications, 2019
- c. Wang, Shuangbao Paul: *“Computer architecture and organization : fundamentals and architecture security”*: Springer, 2021
- d. Godse, Atul P. & Godse, Deepali A.,” *Computer architecture and organization “I*. Technical Publications, 2019
- e. Silberschatz, A. (n,d). Operating System Concepts, 10th Edition John Wiley & Sons, Inc.
- f. Computer Fundamentals - Quick Guide. (n.d.). Retrieved June 20, 2022, from https://www.tutorialspoint.com/computer_fundamentals/computer_quick_guide.htm
- g. Operating System - Process Scheduling. (n.d.). Retrieved June 20, 2022, from https://www.tutorialspoint.com/operating_system/os_process_scheduling.htm
- h. Clive Longbottom Stephen J. Bigelow, (2020) *“Infrastructure (IT infrastructure)”*: Retrieved from <https://www.techtarget.com/searchdatacenter/definition/infrastructure#:~:text=Infrastructure%20is%20the%20foundation%20or,processing%20and%20analysis%20of%20data.>
- i. Priya, B. (2021). What is the operating system evolution? <https://www.tutorialspoint.com/what-is-the-operating-system-evolution>
- j. Technology Platform: from <https://www.sap.com/sea/products/technology-platform/what-is-a-technology-platform.html#:~:text=Technology%20platform%20definition,the%20technology%20that%20supports%20them>

- k. Banck and Front-end Developer from www.coursera.org
- l. What Is an IDE? from <https://www.codecademy.com/article/what-is-an-ide>
- m. Types of Software Platforms from <https://www.geeksforgeeks.org/types-of-software-platforms/>
- n. Drupal vs WordPress – Which CMS Platform is Better For You? from https://www.hostinger.ph/tutorials/drupal-vs-wordpress?ppc_campaign=google_search_generic_hosting_all&bidkw=defaultkeyword
- o. Priya (October 2023) What is Content Management Systems? 8 CMS ypu need to know from <https://themeisle.com/blog/what-is-a-content-management-system-cms/#gref>

MODULE 6

DATABASE CONCEPT, SOFTWARE AND CLOUD CONTAINERS



Introduction

Today's lesson will define the concept of database and software development and define the different database drivers. Just keep in mind our previous topic is about technology platforms. This will help you to differentiate the types of platform technologies.



Learning Outcomes

At the end of this lesson, the student should be able to:

1. identify the different Database drivers' applications;
2. recall how to use server and client connectivity;
3. apply the Front-end and Back-end programming concepts; and
4. discuss the basic cloud infrastructure.

Lesson 1. Database Concepts

What is Database?

- A database is an organized collection of structured information, or data, typically stored electronically in a computer system. A database is usually controlled by a database management system (DBMS), (www.oracle.com/).
- A database is information that is set up for easy access, management and updating. Computer databases typically store aggregations of data records or files that contain information, such as sales transactions, customer data, financials and product information ([/www.techtarget.com/](http://www.techtarget.com/))

What are databases used for?

Businesses use data stored in databases to make informed business decisions. Some of the ways organizations use databases include the following:

- **Improve business processes.** Companies collect data about business processes, such as sales, order processing and customer service. They analyze that data to improve these processes, expand their business and grow revenue.
- **Keep track of customers.** Databases often store information about people, such as customers or users. For example, social media platforms use databases to store user information, such as names, email addresses and user behavior. The data is used to recommend content to users and improve the user experience.
- **Secure personal health information.** Healthcare providers use databases to securely store personal health data to inform and improve patient care.
- **Store personal data.** Databases can also be used to store personal information. For example, personal cloud storage is available for individual users to store media, such as photos, in a managed cloud.

Evolution of databases

Databases were first created in the 1960s. These early databases were network models where each record is related to many primary and secondary records. Hierarchical databases were also among the early models. They have tree schemas with a root directory of records linked to several subdirectories.

Relational databases were developed in the 1970s. Object-oriented databases came next in the 1980s. Today, we use Structured Query Language (SQL), NoSQL and cloud databases.

What are the components of a database?

While the different types of databases vary in schema, data structure and data types most suited to them, they are all comprised of the same five basic components.

1. **Hardware.** This is the physical device that database software runs on. Database hardware includes computers, servers and hard drives.

2. **Software.** Database software or application gives users control of the database. Database management system (DBMS) software is used to manage and control databases.
3. **Data.** This is the raw information that the database stores. Database administrators organize the data to make it more meaningful.
4. **Data access language.** This is the programming language that controls the database. The programming language and the DBMS must work together. One of the most common database languages is SQL.
5. **Procedures.** These rules determine how the database works and how it handles the data.

What are database challenges?

Setting up, operating and maintaining a database has some common challenges, such as the following:

- **Data security** is required because data is a valuable business asset. Protecting data stores requires skilled cybersecurity staff, which can be costly.
- **Data integrity** ensures data is trustworthy. It is not always easy to achieve data integrity because it means restricting access to databases to only those qualified to handle it.
- **Database performance** requires regular database updates and maintenance. Without the proper support, database functionality can decline as the technology supporting the database changes or as the data it contains changes.
- **Database integration** can also be difficult. It can involve integrating data sources from varying types of databases and structures into a single database or into data lakes and data warehouses.

Lesson 2. Types of Databases

There are many different types of databases. The best database for a specific organization depends on how the organization intends to use the data.

- **Relational databases** - A relational database is a type of database that stores and provides access to data points that are related to one another. Relational databases are based on the relational model, an intuitive, straightforward way of representing data in tables.
- **Object-oriented database** - information in an object-oriented database is represented in the form of objects, as in object-oriented programming.
- **Distributed database** - A distributed database consists of two or more files located on different sites. The database may be stored on multiple computers, located in the same physical location, or scattered over different networks.
- **Data warehouse** - A central repository for data, a data warehouse is a type of database specifically designed for fast query and analysis.
- **NoSQL Database** - A NoSQL, or nonrelational database, allows unstructured and semi structured data to be stored and manipulated (in contrast to a relational database, which defines how all data inserted into the database must be composed).
- **Graph Database** - A graph database stores data in terms of entities and the relationships between entities.
- **Open-Source Database** - An open-source database system is one whose source code is open source; such databases could be SQL or NoSQL databases.
- **Cloud Database** - A cloud database is a collection of data, either structured or unstructured, that resides on a private, public, or hybrid cloud computing platform. There are two types of cloud database models: traditional and database as a service (DBaaS).
- **Multimodel Database** - Multimodel databases combine different types of database models into a single, integrated back end. This means they can accommodate various data types.
- **Document/JSON Database** - Designed for storing, retrieving, and managing document-oriented information, document databases are a modern way to store data in JSON format rather than rows and columns.

- **Self-driving Database** - The newest and most groundbreaking type of database, self-driving databases (also known as autonomous databases) are cloud-based and use machine learning to automate database tuning, security, backups, updates, and other routine management tasks traditionally performed by database administrators.

Lesson 3. Front-end and Back-end Development

Front-end Development

- A Front-End Developer is someone who creates websites and web applications.
- The difference between Front-End and Back-End is that Front-End refers to how a web page looks, while back-end refers to how it works.
- You can think of Front-End as the client side and Back-End as the server side.

Responsibilities of Front-end Developer:

- The main responsibility of the Front-End Developer is the User interface.

The basic languages of Front-End Development are:

- HTML
 - HTML stands for Hyper Text Markup Language
 - It is the standard markup language for Web pages
 - HTML elements are represented by <> tags
- CSS
 - CSS stands for Cascading Style Sheets
 - It describes how HTML elements are to be displayed
- JavaScript
 - JavaScript is the Programming Language for the Web.
 - It can update and change both HTML and CSS. Also, it can calculate, manipulate and validate data.

The front-end program, particularly in HTML and CSS, applies the **DOM ➡ document object model**.

What is Document Object Model (DOM)?

- It is a cross-platform and language-independent interface that treats an HTML or XML document as a tree structure wherein each node is an object representing a part of the document.

Framework and libraries are different from each other. The library supports the class, functions, states, and hooks for react js while the framework is the structure of the application interface of JavaScript.

Back-end Development

- Back-end development means working on server-side software, which focuses on everything you can't see on a website. Back-end developers ensure the website performs correctly, focusing on databases, back-end logic, application programming interface (APIs), architecture, and servers.
- The back-end developer writes code that forms the backbone of a website or app

Back-end Developer Task and Responsibilities:

- Build and maintain the website
- Write high-quality code
- Performance quality assurance (QA) Testing
- Assess efficiency and speed
- Troubleshoot and debug
- Train and support

Lesson 4. Cloud Infrastructure

- It is made up of several components, each integrated into a single architecture supporting business operations.

Cloud infrastructure typically includes the following components:

1. **Virtual Machines (VMs):** Virtual machines are virtualized instances of computer systems that run within the cloud infrastructure. They provide the ability to run applications and operating systems on virtualized hardware.

2. Storage: Cloud infrastructure offers various types of storage options, such as object storage, block storage, and file storage. These storage services provide scalable and durable storage for applications and data.

3. Networking: Cloud infrastructure provides networking capabilities to connect and communicate between different resources and services. It includes virtual networks, load balancers, firewalls, and other networking components.

4. Containers: Containers are lightweight, isolated environments that package applications and their dependencies, allowing for efficient deployment and scalability. Cloud infrastructure often includes container services like Docker and Kubernetes for container orchestration.

5. Serverless Computing: Serverless computing allows developers to write and run code without the need to provision or manage servers. Cloud infrastructure platforms provide serverless services, such as AWS Lambda and Azure Functions, that automatically scale and manage the underlying infrastructure.

6. Databases: Cloud infrastructure platforms offer managed database services, such as AWS RDS, Google Cloud SQL, and Azure SQL Database. These services provide scalable and highly available databases for storing and managing data.

8. Management Tools: Cloud infrastructure platforms provide management tools and APIs to provision, monitor, and manage the resources and services deployed in the cloud. These tools enable automation, monitoring, and configuration management of the cloud infrastructure.

9. Security and Compliance: Cloud infrastructure providers implement robust security measures to protect data and resources. They adhere to industry-standard security practices and regulatory compliance requirements.



Assessment Tasks

Instruction: Read the scenario, reflect, and give your opinion on the space provided.

1. Mecca was building a social media platform that allows users to post and share content, follow other users, and engage in discussions. As the user base grows, Mecca realizes that the traditional relational database management system (RDBMS) is struggling to handle the increasing volume of data and the dynamic nature of the application. To address this problem, what type of database that Mecca need to use and why?
2. Geff developed a multimedia content management system that handles a wide range of media types, such as images, videos, and audio files. What type of databases he will use? Why?



Summary

Databases are used for storing, maintaining and accessing any sort of data. They collect information on people, places or things. That information is gathered in one place so that it can be observed and analyzed. Databases can be thought of as an organized collection of information (/www.techtarget.com/).



References

- a. What Is a Database? from <https://www.oracle.com/database/what-is-database/>
- b. Ben Lutkevich and WriterAdam Hughes (February 2023) What is a database? from <https://www.techtarget.com/searchdatamanagement/definition/database>
- c. Banck and Front-end Developer from www.coursera.org

MODULE 7

TYPES OF SOFTWARE PLATFORMS

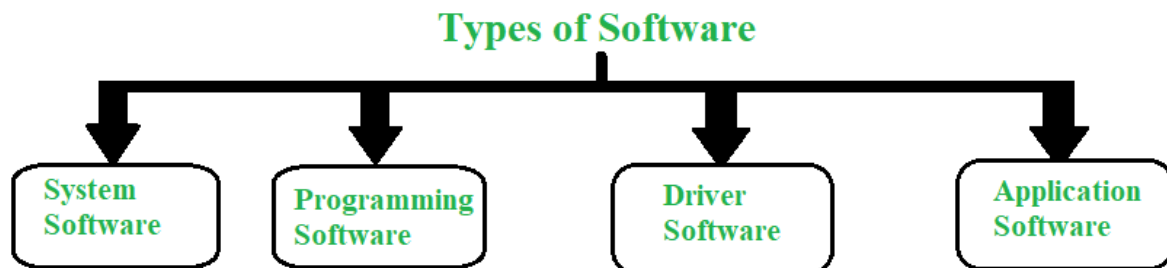


Introduction

Almost every software entrepreneur nowadays builds some kind of a 'platform'. Yet very few will confidently answer a seemingly simple, but very important question: What kind of platform do you build? All platforms are not created equal. Google Search, Facebook, Amazon Web Services, Amazon Marketplace, Android, Uber, AirBnB, Waze, WeWork, Twilio and even Bitcoin are all platforms. At the same time, these platforms are very different in how they create network effects, interactions they enable, approaches to solving "chicken and egg" problems (do you build the demand side first or the supply side?), openness levels, growth dynamics, subsidies, competitive strategies and monetization methods (<https://medium.com/>).

Building a successful platform is more about making the right trade-offs than it is about best technology. To understand these tradeoffs, you must have a good grasp on what kind of platform you are building.

Platform is hardware and software architecture that acts as foundation or base upon which other applications, processes, or technologies are developed. In computing, platform refers to basic hardware i.e., computer system and software i.e., operating system on which software applications are often run. An application also can serve as platform if it's base for other programs. For instance, the web browsers that we use in our day-to-day life accept few third-party plugins, and hence browser application becomes platform for interfacing (<https://www.geeksforgeeks.org/>).



In recent times, almost every software enterprise builds some quite 'platform'. Yet all platforms that are created aren't same. Facebook, Amazon Web Services, Amazon Marketplace, Google Search, Android, Uber are all platforms, but at same time, these platforms are very different in how they create network effects, kind of interactions they allow, approaches they follow, strategies and other methods. Now let us discuss the various types of platforms. They are: Utility Platforms; Content Distribution Platforms; Data Harvesting Platforms; Interaction Networks; Technology Platforms; Marketplace; Computing Platforms and Content Crowdsourcing Platforms (<https://www.geeksforgeeks.org/>).



Learning Outcomes

At the end of this lesson, the student should be able to:

1. Discuss what is utility platform is?
2. Identify utility platforms can be found in various domains

Lesson 1. Utility Platform

In today's digital age, organizations across various industries are constantly seeking ways to enhance productivity, and efficiency, and streamline their operations. This is where utility platforms come into play. Utility platforms are digital systems or software solutions designed to provide various utility services and functions, catering to specific industry needs.

Utility platforms leverage cutting-edge technologies such as cloud computing, data analytics, artificial intelligence, and automation to deliver their services. By harnessing the power of these technologies, utility platforms offer organizations a centralized and streamlined approach to managing their processes, data, and resources (<https://www.geeksforgeeks.org/>)

Lesson 1.1. What is utility platform?

- It is a software or technology infrastructure that provides various services and functionalities to users. It typically serves as a centralized hub for accessing and managing different utilities or resources.

- Utility Platforms attract their users by providing useful and also usually free services.

-

Lesson 1.2. Utility platforms can be found in various domains such as:

1. Cloud computing

- Utility platforms in cloud computing offer resources such as computing power, storage, and networking on-demand, allowing users to scale their infrastructure based on their needs.

Examples of utility platforms in cloud computing include

- Amazon Web Services
- Microsoft Azure
- Google Cloud Platform



2. Energy Management

- Utility platforms in the energy sector help users monitor, control, and optimize their energy consumption.

Examples of utility platforms in energy platforms include

- Smart grids
- Sensors
- Devices to provide real-time data and analytics for efficient energy management.



3. Internet of Things (IoT)

- IoT utility platforms enable users to connect, manage, and control a network of IoT devices.

These platforms offer features like

- Device provisioning
- Data collection

- Remote monitoring
- Device management
- Facilitating the development and deployment of IoT solutions



4. Smart Cities

- These platforms integrate data from multiple sources, such as transportation systems, public utilities, and civic services, to improve efficiency, sustainability, and quality of life in cities.
- Utility platforms in the context of smart cities provide a centralized system for managing various urban services.

5. Data and Analytics

- Utility platforms focused on data and analytics offer tools and services for processing, analyzing, and visualizing data.

These platforms offer features like:

- Data integration
- Data transformation
- Machine learning algorithms
- Reporting capabilities



6. Communication and Collaboration Platforms

- Utility platforms in the context of smart cities provide a centralized system for managing various urban services.

These platforms offer features like:

- Messaging
- Video Conferencing
- Document Sharing
- Task Management
- Project Collaboration



Example:

- Slack
- Microsoft Teams
- Google Workplace

7. Financial Utility Platforms

- These platforms provide financial services and infrastructure to businesses and individuals.
- They may include payment processing, banking services, financial planning tools, investment platforms, and cryptocurrency exchanges.



A general illustration of the business model for software and hardware requirements:

1. **Identify Target Market:** The first step is to identify the target market for the software or hardware product. This involves understanding the needs and preferences of potential customers and determining the industries or sectors that can benefit from the product.
2. **Market Research and Analysis:** Conduct market research to gather insights into customer needs, preferences, and existing competition. Analyze the market trends, potential demand, and pricing dynamics to inform product development and marketing strategies.
3. **Product Development:** Based on the identified requirements, develop the software or hardware product that aligns with the market needs. This involves designing and building the product, ensuring it meets quality standards, and conducting thorough testing to

ensure functionality and reliability.

4. **Pricing Strategy:** Determine the pricing strategy for the software or hardware product. Consider factors such as production costs, competitive pricing, perceived value, and market demand to set an optimal price point that balances profitability and customer acceptance.
5. **Distribution Channels:** Define the distribution channels through which the software or hardware product will reach the target market. This can include direct sales, partnerships with retailers, online platforms, or a combination of channels. Develop relationships with channel partners and establish agreements to ensure effective product distribution.
6. **Sales and Marketing:** Develop a comprehensive sales and marketing strategy to promote the software or hardware product to the target market. This may involve creating marketing collateral, attending industry events, utilizing digital marketing channels, and building a strong online presence. Engage in targeted advertising, lead generation, and customer relationship management to drive sales.
7. **Customer Support:** Establish customer support mechanisms to address inquiries, and technical issues, and provide assistance to customers. This can include online documentation, FAQs, troubleshooting guides, and responsive customer service through various channels such as phone, email, or chat.
8. **Revenue Generation:** Generate revenue through product sales or subscription models. For software, revenue models may include one-time purchases, recurring subscriptions, or usage-based pricing. For hardware, revenue can come from direct sales or leasing/rental options.
9. **Continuous Improvement:** Gather customer feedback and market insights to continuously improve the software or hardware product. Release updates, bug fixes, and new features to address customer needs, maintain competitiveness, and enhance customer satisfaction.
10. **Partnerships and Expansion:** Explore partnerships and collaborations to expand the reach of the software or hardware product. This may involve partnering with other companies to integrate the product into larger solutions or entering new markets through strategic alliances or acquisitions.

11. **Financial Management:** Ensure effective financial management by tracking revenue, expenses, and profitability. Monitor key performance indicators (KPIs) such as sales growth, and customer acquisition costs, and customer lifetime value to make informed business decisions and optimize financial performance.

Lesson 2. Content Distribution Platforms

To effectively reach your intended audience, consider using content distribution platforms. It's important to ensure that your content is engaging and relevant, and provides actionable insights to achieve the best possible results.

What Is Content Distribution?

Content distribution means disseminating content to select target audiences through various media formats and channels. There are three categories under which all these content distribution platforms fall.

Paid media comprises the content distribution channels you need to pay for to place your ads, whether social media or pay-per-click (PPC) advertising. Paying for your ads is a quick way for your brand to get exposure. Owned media comprises content distribution channels you have absolute control over. They are essentially property you own; examples include websites, email, and blogs. Earned media relies on other people for distributing content and includes product reviews, guest posts, testimonials, etc. This is the most organic way of content distribution, and it increases your brand's credibility tenfold.

Choosing a web content distribution platform that best suits your brand's needs will help you boost your distribution efforts. Without the right content distribution platforms, it is impossible to promote your content to many people who make up your target audience (www.peppercontent.io/blog/best-content-distribution-platforms/)

Lesson 2.1 What is content distribution platform?

As a user, it's important to understand the significance of content distribution and the role it plays in promoting your content to a new audience.

What is Content Distribution?

Content distribution involves disseminating content to select target audiences through various media formats and channels.



A content distribution platform is a website dedicated specifically to promoting user-submitted content. Unlike display ads, it can encompass native advertising.

Google AdSense, PropellerAds, and Millennial Media are samples of **Content Distribution Platforms**. Such platforms connect owners with content who are wishing to deliver content (or ads) to users. More content available on platform, more attractive platform becomes. User reach and accuracy of content matching are two most important aspects of this platform.

So, before we dive into the eleven best content distribution platforms, let's remember why promoting content is crucial. After all, you spend hours creating a blog post, researching, and writing it flawlessly with no plagiarism. Promoting your content is just as important as creating it, which is why having a good content distribution strategy and utilizing the best content distribution platforms is essential.

Lesson 2.2 Best content distribution platform

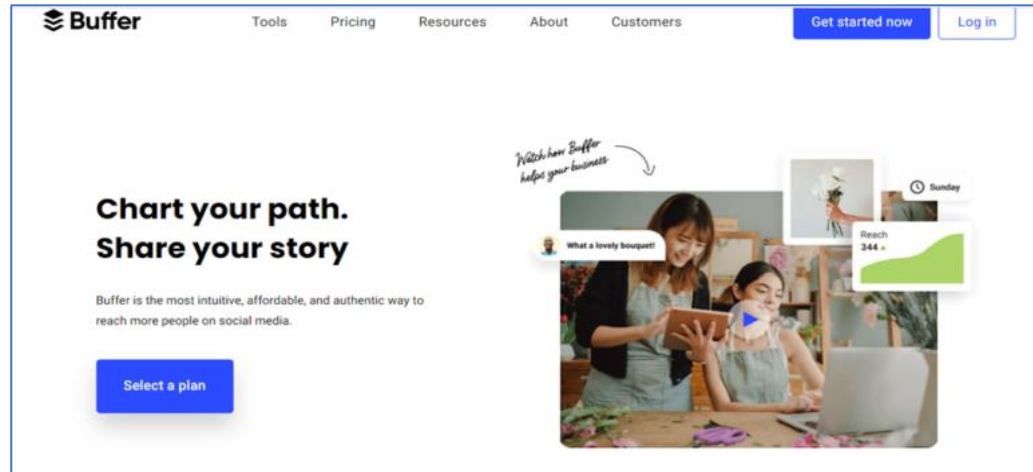
(www.peppercontent.io/blog/best-content-distribution-platforms/)

The essence of content promotion and distribution is sharing content that's informative, valuable, unique, relevant, and, most importantly, actionable. Your content should be viewed by more people online and by people looking for the same content, with the hope that it offers the best solution. Here's a handy list of content distribution tools that will help boost your web content distribution efforts:

1. Buffer

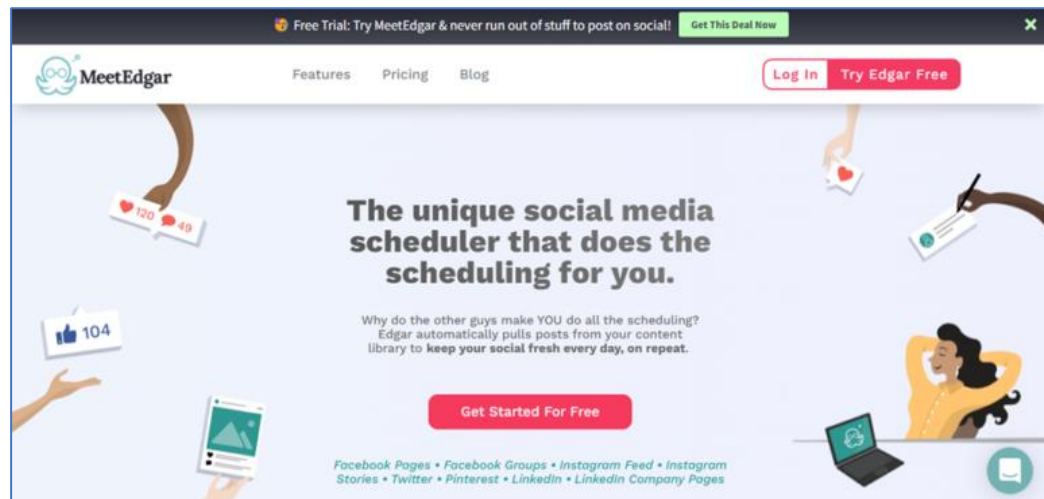
It is one of the best content distribution tools that provide the simplest mean. All it takes is one click to accomplish the task. What's more, you may have the liberty to have the

content queued and scheduled to appear at specific times. Adding images or videos makes your old content new, and gives Buffer a fresh lease of life. You can do this right from the Buffer dashboard. You can measure your metrics using Buffer analytics and tweak your campaigns when needed.



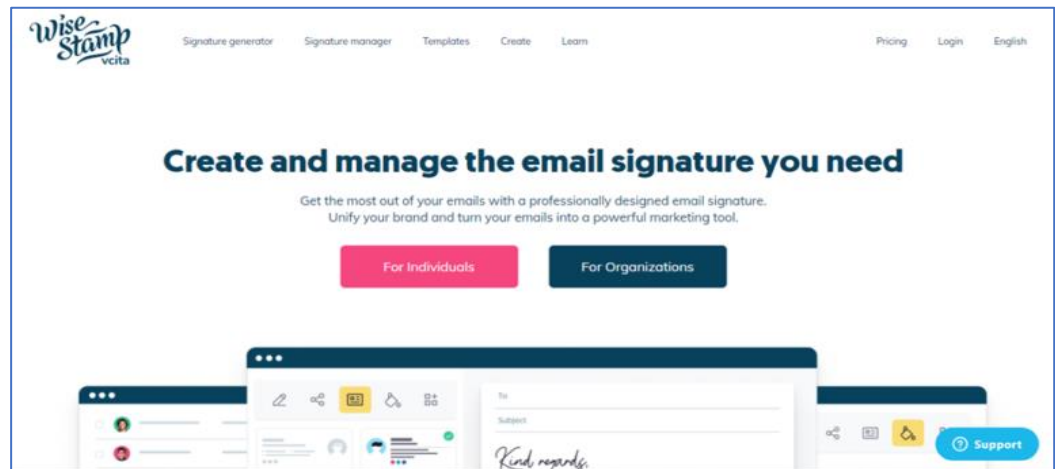
2. MeetEdgar

An efficient web content distribution platform, MeetEdgar, is quick and easy. It comes to your aid when your team is running short on content. It not only helps you save precious time you would have spent on creating new content, but it also ensures you are continued presence on social media, which is essential to remain fresh in the minds of your target audiences. You can even schedule posts for upload with this tool.



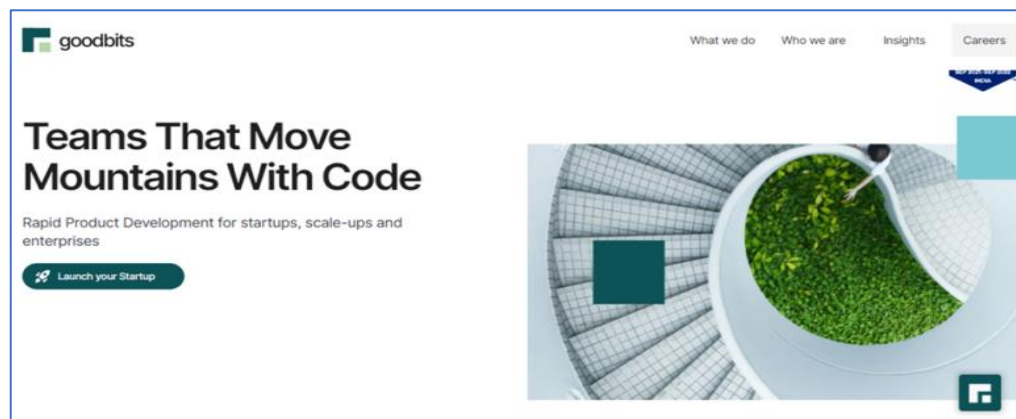
3. WiseStamp

If your business demands that you write more messages, WiseStamp is one of the best content distribution tools for the job. Besides sharing your information, you can also share your social media account details and email signature. You can leave a call to action (CTA) button as a link to your Twitter handle. You can have CTAs at the end of your blog posts and link them via RSS feeds.



4. Goodbits

It is another content distribution tool that lets you share your content with your target audience via email newsletters. You can leverage the long hours spent on content creation by automating content from RSS feeds, Twitter, and all the content posted on Pocket, Slack, and the ubiquitous Facebook. You can easily collect content, and you'll



love the drag-and-drop editor that lets you manually collect information. You can also customize your mailing list when using Goodbits.

5. Mailchimp

Mailchimp lets you manage your subscribers and helps send email newsletters to every user in your contact list. What's more, you can track and monitor the results of your campaign using this content distribution channel. You can use it to automate and distribute all your new posts. And you can rely on your analytics reports to track all emails sent as part of a campaign and apply the data to enhance your future campaigns.



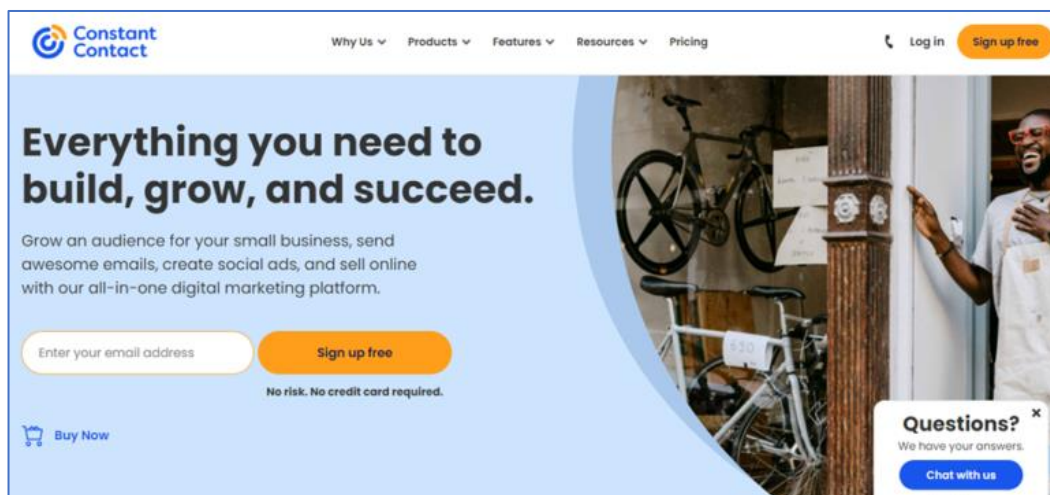
6. SumoMe

It is a WordPress plugin with several tools that come in handy for distributing earned and owned media. SumoMe has excellent content distribution tools for owned media, such as giveaway widgets and a sign-up bar, which helps create your email lists. For earned media, there is a tool known as "Share," which helps track users responding to your ads and helps redistribute your content through more than a dozen social media networks.



7. Constant Contact

As the name signifies, this content distribution tool helps you remain in constant touch with your audience through email distribution. The unique drag-and-drop feature lets you upload images and other branding content to email templates and share them with your customers. The email sign-in form helps you gain new customers through any of your owned media properties. You can automatically transform emails into social media posts using Constant Contact and build your own social media communities to increase your reach.



8. Click to tweet

This is a handy content distribution tool as it lets you share your content, and you can also create links to all your posts and announce product launches. As soon as the customer clicks on the link, your original message is automatically transferred to the Twitter compose page. It makes it a lot easier for your members on Twitter to share your information without having to type. This easy feature makes it possible for your launch to go viral on social media, especially Twitter, and spread the news to more people in one go.

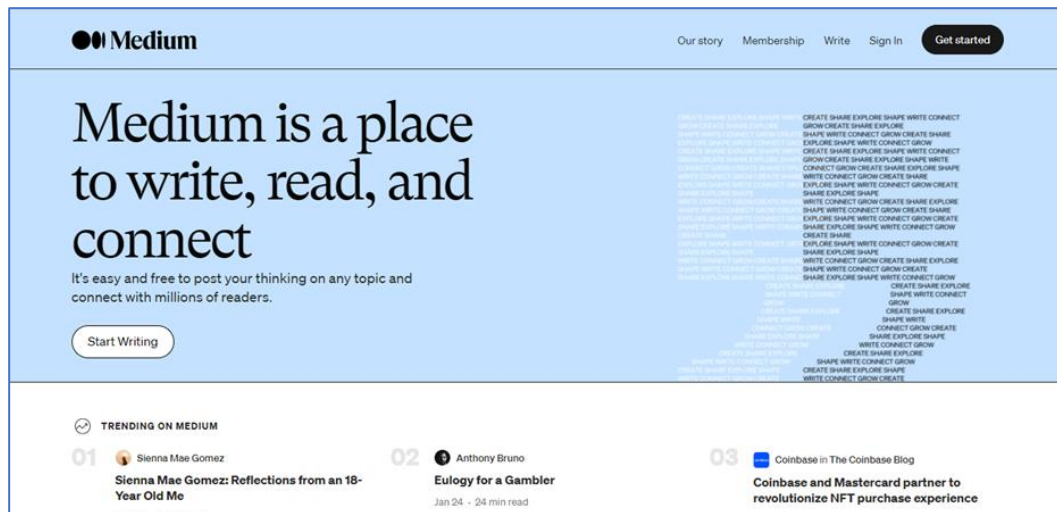


9. Medium

The co-founder of Twitter created Medium. This efficient content distribution platform helps in paid and earned content distribution with no limitation on the length of posts. Medium is used by over 80 million people worldwide. Having access to this active audience is gold for content distributors.

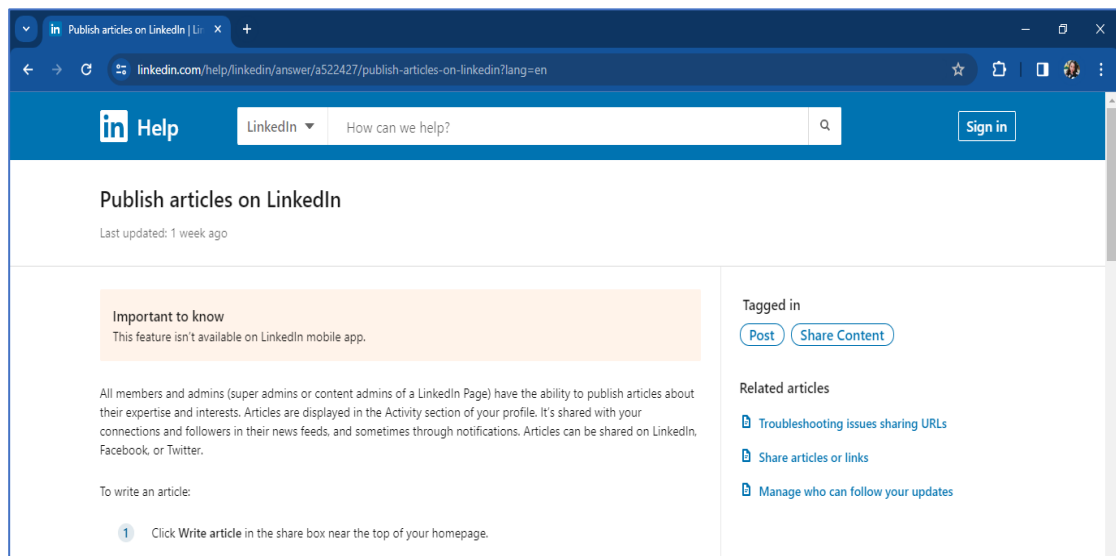
The atmosphere on Medium is relaxed with no need to rush things, and it pays to use in-depth content that's both engaging and informative. Your posts should match the mood

on this network. Popular multinational companies using mediums include Starbucks, BMW, Slack, Intel, and more.



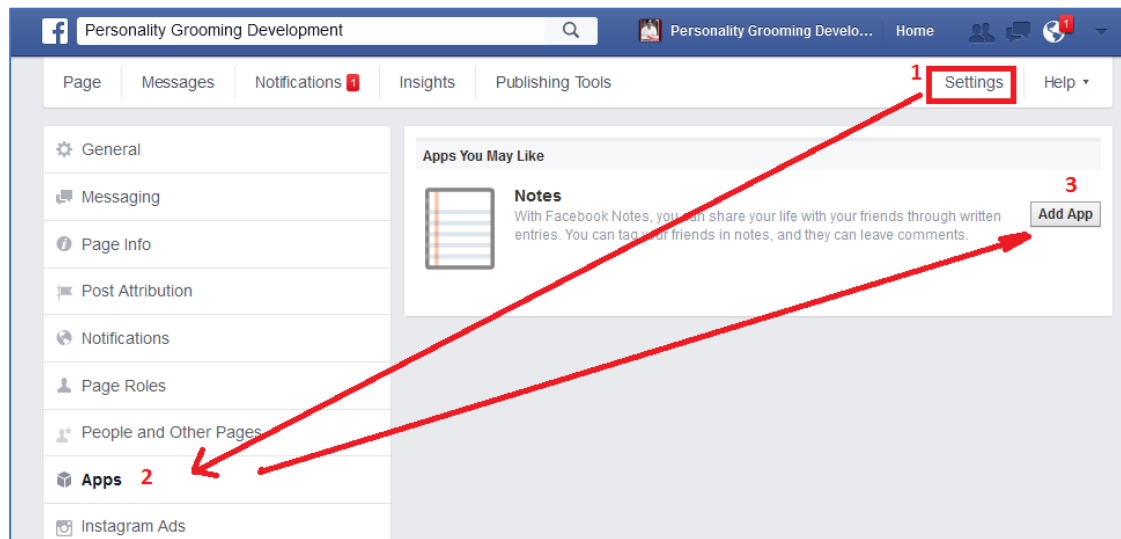
10. LinkedIn Publisher

LinkedIn Publisher is one of the handiest content distribution tools that helps a brand reach more users. LinkedIn is a platform frequented by professionals, and relevant content is constantly viewed and re-shared promptly. Many posts have gone viral on LinkedIn and shared on other social media platforms from there. The homepage features a “Publish a Post” button, and clicking on the button lets you upload images and content.



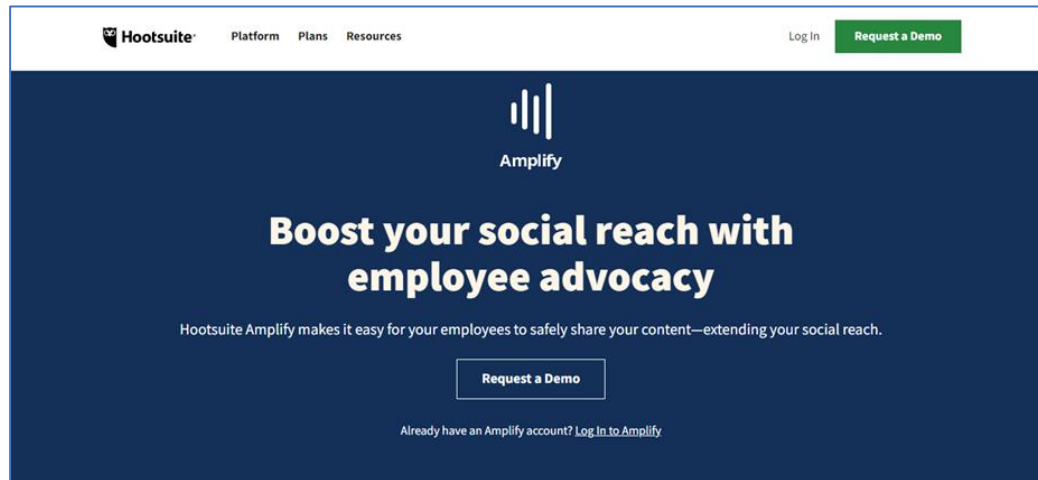
11. Facebook Notes

Facebook's Notes feature lets you publish text and images. You can post content from your profile or business page. You can summarize a long blog post that readers might skip and share it as a note. Each of these notes will link back to the blog post, giving your readers the chance to read the entire blog on your website. It is easy to find Notes on Facebook: simply go to your profile and look for the "More" tab. As well as promoting your business or personal profile, you can drive traffic to your website with Notes. It is a helpful content distribution tool. All you need to do is link your Note to the landing page on your website.



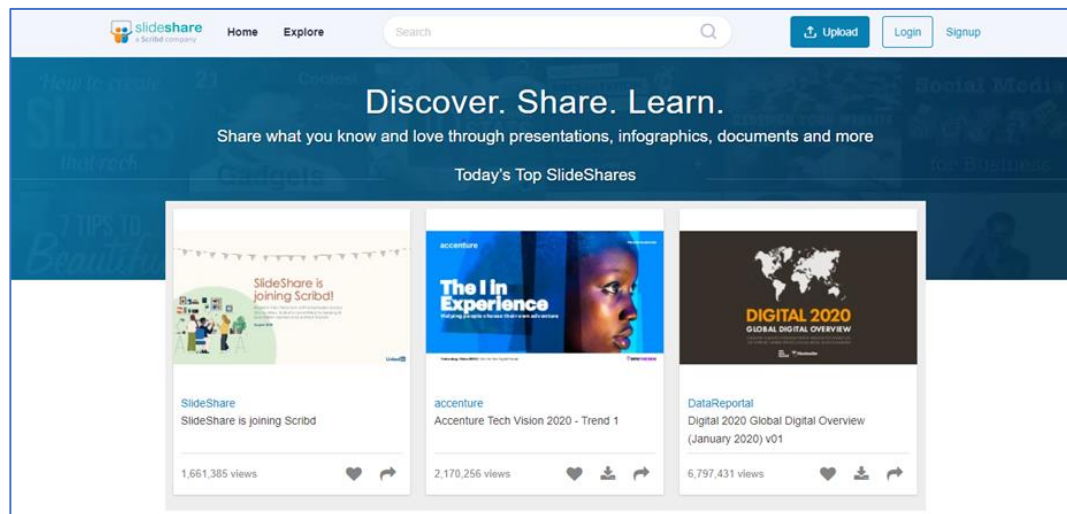
12. Hootsuite Amplify

As the name suggests, this content distribution tool lets you amplify your content by pushing notifications about fresh content to your associates, colleagues, and employees, giving them a chance to share it with their peers through their social media accounts. This tool is ideal for companies with a large employee base, as it helps with distributing content more widely and in one go.



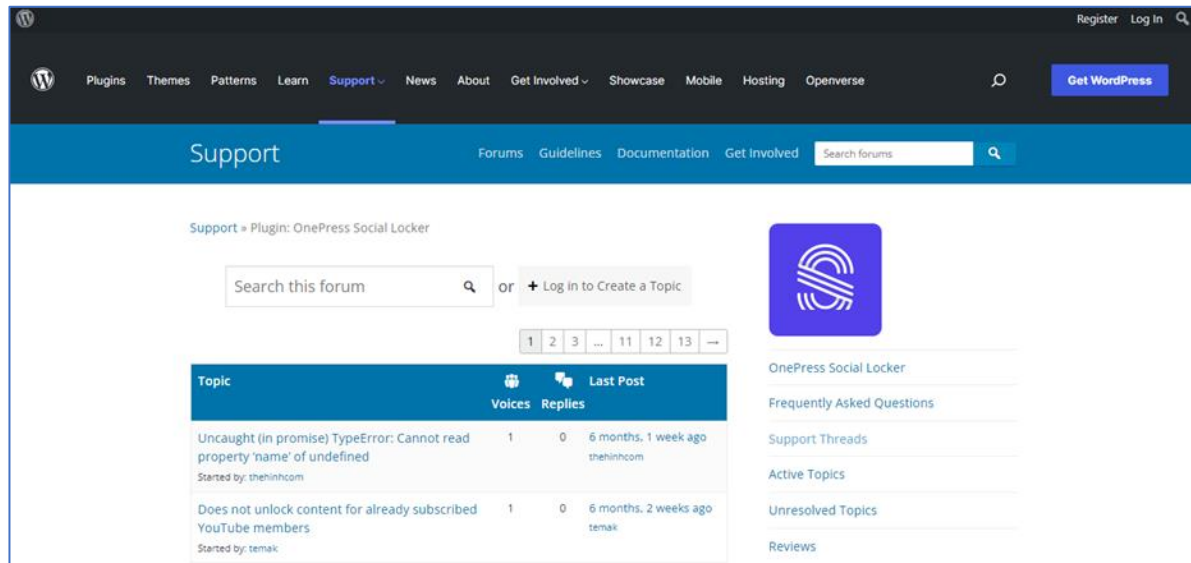
13. SlideShare

[SlideShare](#) is a handy content distribution tool that lets you share content as a series of slides (slideshow). With millions of users worldwide, your content has excellent potential to reach a large audience base. Besides PowerPoint presentations, you can also share your videos and infographics through the same tool.



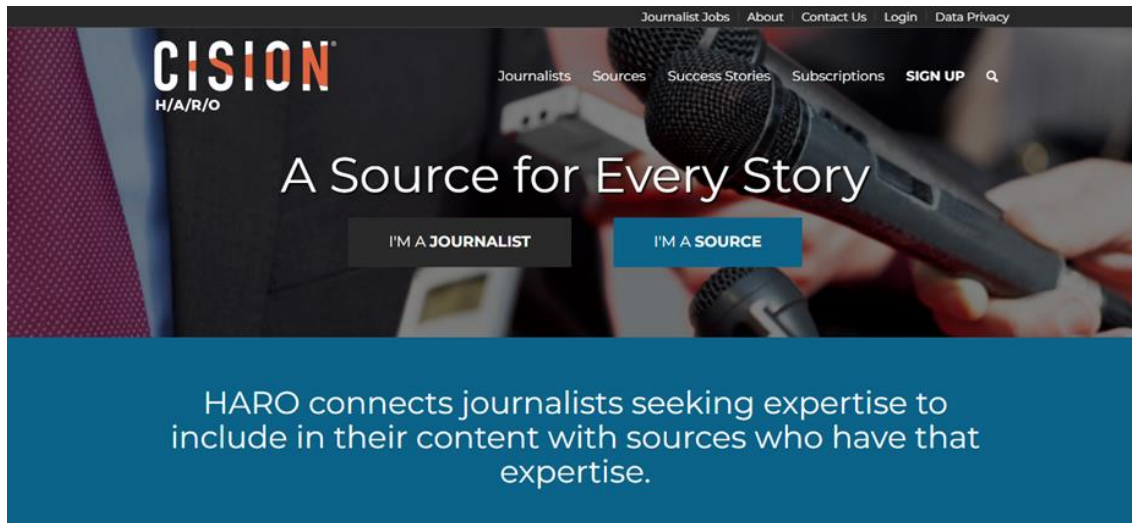
14. OnePress Social Locker

With this tool, you can control what content people can see. More specifically, only the later-published content. If you use this content distribution tool, you can lock any social media content so that readers can only access it after they have shared it on Twitter and Facebook.



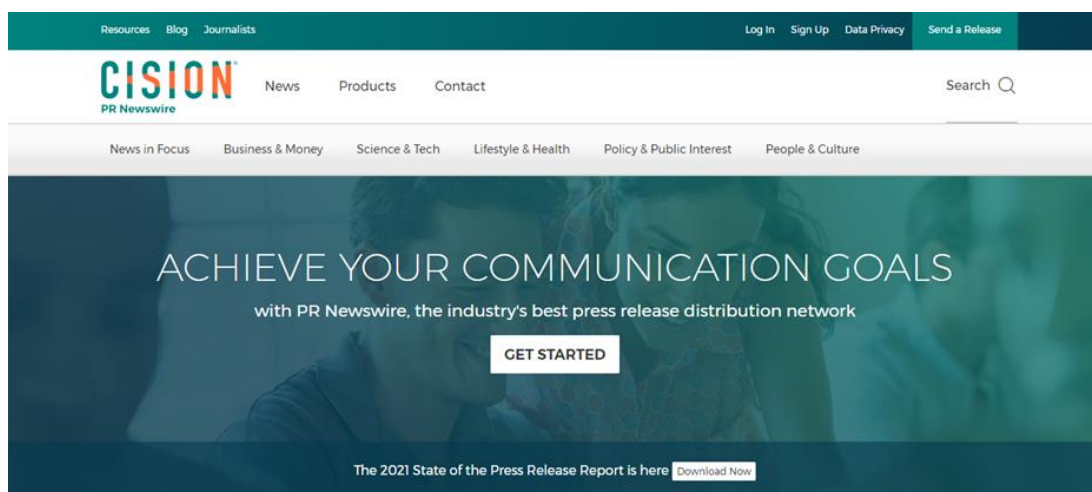
15. Help a Reporter Out (HARO)

HARO is one of the most effective content distribution platforms around the world. HARO is used to share more than 50,000 articles every year on leading publications such as Reuters, Chicago Tribune, The New York Times, etc. It presents an excellent opportunity for content distributors to locate relevant topics related to their domain and contact a journalist for distribution. It is highly effective to establish a relationship with such writers and promote your website through the articles published by such big media houses.



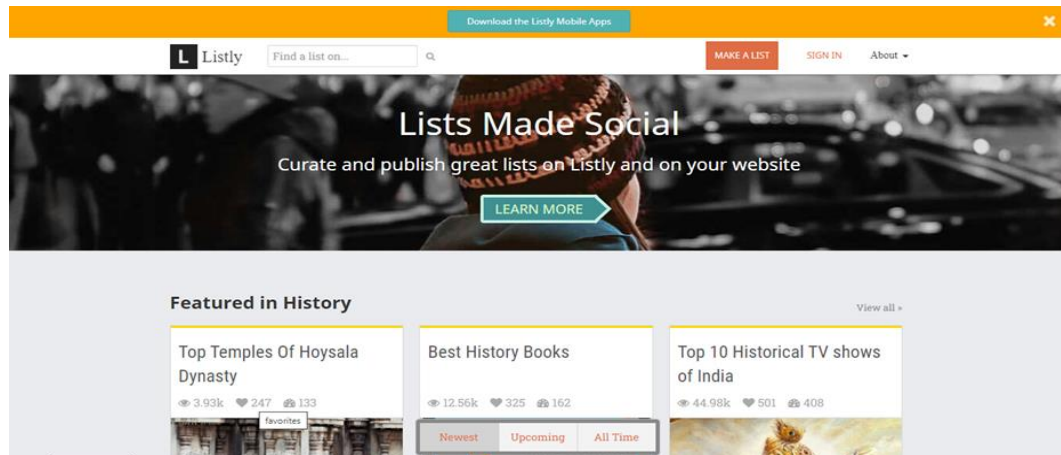
16. PR Newswire

As the name suggests, this content distribution tool is an excellent means to share newsworthy content, including exciting announcements, upcoming events, breaking news, and any other important information regarding your brand, to various sources. You need to sign up on [PR Newswire](#) and have your account authorized to assist your promotions. There's excellent potential to distribute your content on close to 200,000 media sources, and this tool is one of the handiest to promote content about your company and brand.



17. Listly

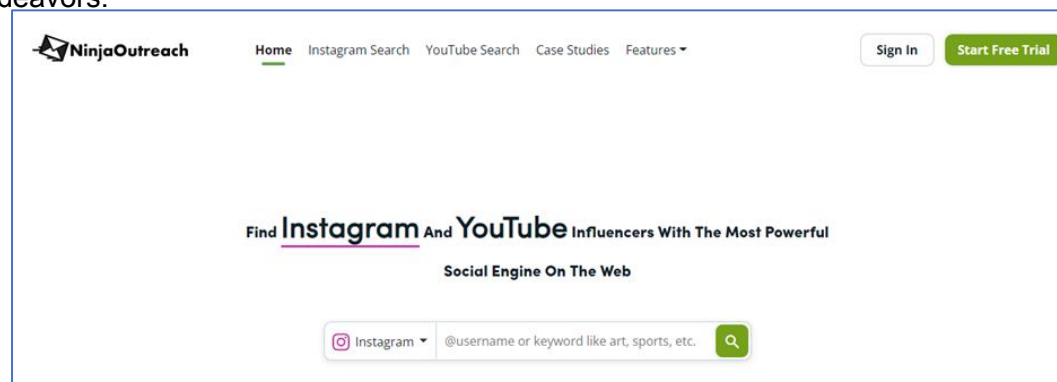
This is an easy-to-use content distribution tool, especially if you need to build lists in a specific area and publish them. Listicles are a popular form of content as the format is easily scannable, and sharing it is also easy. In addition to building lists related to your niche, this tool is also helpful in getting your content the visibility it needs.



18. NinjaOutreach

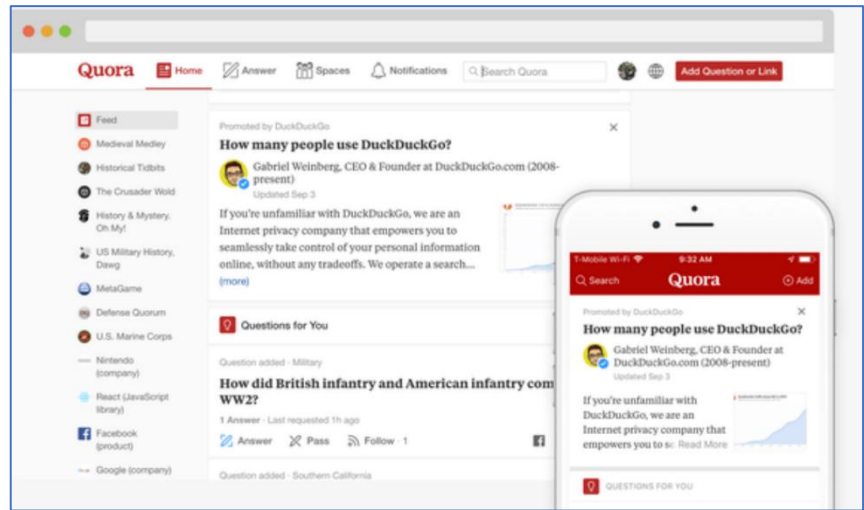
Influencers have a way with their followers and help cement relationships with visitors on your website. They help share your content with a large following, and the chances of publishing as a guest blogger also exist. NinjaOutreach is an excellent content distribution tool for popularizing and promoting your content through influencers.

This content distribution tool can capture commonly used phrases to locate influencers, make convenient lists, and create emails using inbuilt templates. You can send them to recipients in your contact list within seconds, thus eliminating the need for more complicated spreadsheets or multiple tools for managing your outreach endeavors.



19. Quora

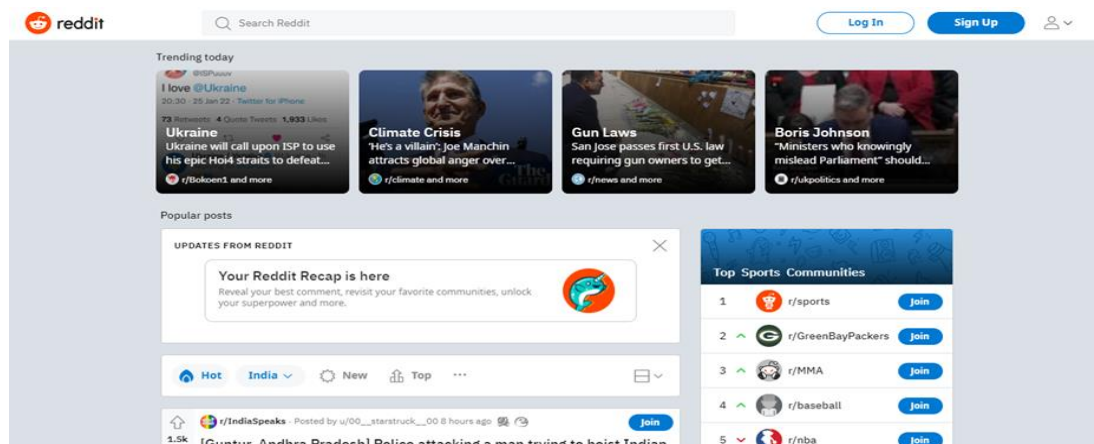
This content distribution platform helps you to share your brand expertise in the form of answering questions. The best place to ask questions and find a range of answers



is Quora. You can ask questions on any topic on the planet and have experts give short, detailed, and informative answers. You can also share your knowledge on Quora, which helps build your online community. This content distribution platform enables you to share your brand expertise by answering questions.

20. Reddit

Reddit is the web content distribution platform where most people go when searching for informative content on any topic on the internet. The posts are shared through subreddits, exclusive communities that exchange notes on industry-related topics. Find a subreddit relevant to your domain, exchange notes with like-minded people, and share content. The best content is regularly voted and appears on the top of the pages, leading to more shares.





Assessment Tasks

Activity No. 1: Instruction: Read the questions carefully below. Write **T** if the statement is **true** and **F** if the statement is **false** put your answer in the space provided before the number.

- _____ 1. Identifying the target market involves understanding the wants and needs of the potential customer.
- _____ 2. Data and analysis is one of the utility platforms in the context of smart cities that provide a centralized system for managing various urban services.
- _____ 3. Gathering customer feedback and market insights will help the software and hardware products improve.
- _____ 4. Payment processing, banking services, financial planning tools, investment platforms, and cryptocurrency exchanges were included in the financial utility platform.
- _____ 5. A utility platform is a software or technology infrastructure that provides various services and functionalities to users.
- _____ 6. IoT utility platforms enable users to connect, manage, and control a network of IoT devices.
- _____ 7. Financial utility platforms in the energy sector help users monitor, control, and optimize their energy consumption.
- _____ 8. Distribution Channels was involve creating marketing collateral, attending industry events, utilizing digital marketing channels, and building a strong online
- _____ 9. In computing, platform refers to basic software i.e., computer system and hardware i.e., operating system on which software applications are often run.

_____10. Utility platforms in the context of smart cities provide a centralized system for managing various urban services.

Activity No. 2: Instruction: Read the questions carefully and write the correct answer in the space provided below.

_____1. It is one of the handiest content distribution tools that helps a brand reach more users.

_____2. It is one of the best content distribution tools that provide the simplest mean.

_____3. It is a crucial aspect of content marketing as it is the key to boosting your followers count.

_____4. This content distribution tool helps you remain in constant touch with your audience through email distribution.

_____5. With this content distribution, you manage your subscribers and help send email newsletters to every user in your contact list. What's more, you can track and monitor the results of your campaign using this content distribution channel.



Technology Platforms: Amazon Web Services, Microsoft Azure, and Twilio are examples of Technology Platforms. Technology Platforms provide building blocks or services that are reused in a large number of products. Through permission-less innovation 3rd party developers embed these building blocks and services in their products, driving more adoption of the platform.

Google Search, Kayak and Zenefits are examples of Utility Platforms. Utility Platforms attract users by providing a useful, typically free service. Once there is critical mass of users using the service, the platform opens to the second type of participants, advertisers in the case of Google Search, airlines in the case of Kayak or insurance companies in the case of Zenefits. There is no network effect in the useful service itself. Users attract businesses, but businesses on the platform do not necessarily attract users. We go to Google Search looking for information, not to see ads.

Launching Utility Platforms is fairly straightforward — Make sure you have a useful service that generates repeated use and has negligible marginal cost. Once you have an asset created by a critical mass of users (e.g. data, targeted engagement, etc.), open your service to businesses to monetized the platform (for example, through advertising, commissions or anonymized data).

Key takeaways on why we need to know the best content distribution platforms:

- Content distribution is a crucial aspect of content marketing as it is the key to boosting your follower's count.
- Content distribution directly impacts brand awareness, in turn exposing your content on social media platforms and fostering engagement.
- Content distribution channels can be categorized into three groups: Paid, Owned, and Earned Media.
- Paid content distribution platforms, like its name, distribute content for explicit pay. A pay-per-click model is in place here.
- Owned content distribution tools are mostly in-house content like blogs, email newsletters, social media posts, and other products distributed on the host website or other platforms.
- Earned content distribution channels mean distributing content through third parties like reviews, retweets, press coverage, guest postings, etc.
- Edgar, Wise Stamp, Goodbits, and SumoMe are content distribution tools to promote your content on multiple platforms.
- You can use automated content distribution to promote your content by automating your email newsletters. Then, you have to connect it to content marketing platforms to

distribute.

- Web content distribution means promoting your content to online users in multiple formats. Bite-sized video content is the most popular form, followed by gifs and static pictures.
- Brands need content distribution platforms, as users need to find and engage with your content.
- Some of the best content distribution platforms include Buffer, MeetEdgar, Mailchimp, WiseStamp, Reddit, Quora, Facebook, Twitter, etc.



References

- a. Team Pepper (2021): *"36 Best Content Distribution Platforms To Know About"* from <https://www.peppercontent.io/blog/best-content-distribution-platforms/>
- b. *Types of Software Platforms* from <https://www.geeksforgeeks.org/types-of-software-platforms/>

- END OF MODULE FOR MIDTERM PERIOD –
END OF THE MIDTERM MODULE CHECK YOUR EXAM SCHEDULE
DO NOT FORGET TO TAKE THE EXAM AS SCHEDULED